

Optimizing the Expansion of Burn Care Access: A Facility Location Model with Coverage and Equity

Team 17

March 16, 2026

Abstract

Burn injuries require specialized care that is often concentrated in designated centers. In the United States, many patients must travel long distances to reach a burn center, and access is uneven across geographic and demographic groups. We develop a mixed-integer linear programming model to identify hospitals that should be upgraded to adult or pediatric burn centers to improve access equity. The model minimizes population-weighted travel distance while ensuring that a target fraction of the population is within a threshold distance (coverage) and limiting the number of new upgrades. Using nationwide ZCTA-level population data and a comprehensive hospital dataset, we evaluate the impact of adding burn capabilities to selected hospitals. Results show that upgrading a modest number of hospitals can dramatically reduce average and maximum travel distances, increase coverage, and reduce disparities. Our approach provides a data-driven tool for policy makers and health planners to allocate resources equitably.

1 Introduction

Timely access to specialized burn care is critical for patient outcomes, yet many regions of the United States are underserved. The American Burn Association verifies only 76 burn centers, and eight states have none at all [1]. Furthermore, over half of burn patients are treated at non-verified centers [3]. The lack of a systematic method to identify where to add burn care capacity hinders efforts to improve access. This study addresses that gap by formulating an optimization model that recommends hospitals for upgrading to adult or pediatric burn centers, balancing efficiency (total travel distance) with equity (coverage within a threshold).

2 Data Sources

2.1 Demand Estimation

We used ZCTA (ZIP Code Tabulation Area) level population data from the 2023 American Community Survey 5-year estimates (Table B01001). For each ZCTA i , we computed adult

population (age ≥ 18) and pediatric population (age ≤ 18). Annual burn care demand was estimated by applying a national incidence rate of 600,000 burn injuries per year [2]:

$$h_i^{\text{adult}} = \frac{\text{adult}_i}{\text{total US pop}} \times 600,000,$$

$$h_i^{\text{ped}} = \frac{\text{under18}_i}{\text{total US pop}} \times 600,000.$$

ZCTA centroids (latitude/longitude) were obtained from the Census TIGER/Line shapefiles.

2.2 Hospital Data

A comprehensive list of hospitals with their locations and current burn care capabilities was compiled from the National Injury Resource Database (NIRD) [1], supplemented by the American Burn Association, American College of Surgeons, and state health departments. For each hospital j we recorded:

- Existing adult burn care: `burn_adult` (0/1)
- Existing pediatric burn care: `burn_peds` (0/1)

- Latitude and longitude (geocoded).

The final dataset contains 635 unique hospitals across the U.S.

2.3 Distance Matrix

Travel distances between each ZCTA centroid and each hospital were computed using the Haversine great-circle formula, yielding a matrix d_{ij} in kilometers. For computational efficiency, the matrix was stored as a dense NumPy array.

3 Mathematical Model

We formulate a mixed-integer linear program (MILP) to select hospitals for upgrading. Let:

- I = set of demand points (ZCTAs), indexed by i
- J = set of candidate hospitals, indexed by j
- $K = \{A, P\}$ = care types (adult, pediatric)
- h_i^k = annual burn demand of type k at ZCTA i
- d_{ij} = distance (km) from ZCTA i to hospital j
- $\text{exist}_j^k = 1$ if hospital j already provides type k care
- T = coverage threshold (km)
- α = minimum fraction of demand that must be covered within T
- p^k = maximum number of *new* hospitals to be designated for type k

Decision variables:

- $x_j^k \in \{0, 1\}$: 1 if hospital j provides type k after expansion
- $y_{ij}^k \in \{0, 1\}$: 1 if ZCTA i is assigned to hospital j for type k

Objective:

$$\min \sum_{k \in K} \sum_{i \in I} \sum_{j \in J} h_i^k d_{ij} y_{ij}^k.$$

Constraints:

$$\sum_{j \in J} y_{ij}^k = 1 \quad \forall i \in I, k \in K \quad (1)$$

$$y_{ij}^k \leq x_j^k \quad \forall i \in I, j \in J, k \in K \quad (2)$$

$$x_j^k \geq \text{exist}_j^k \quad \forall j \in J, k \in K \quad (3)$$

$$\sum_{j \in J} (x_j^k - \text{exist}_j^k) \leq p^k \quad \forall k \in K \quad (4)$$

$$\sum_{i \in I} \sum_{j \in J: d_{ij} \leq T} h_i^k y_{ij}^k \geq \alpha \sum_{i \in I} h_i^k \quad \forall k \in K \quad (5)$$

$$x_j^k, y_{ij}^k \in \{0, 1\} \quad \forall i, j, k \quad (6)$$

Constraint (1) ensures every ZCTA is assigned to exactly one hospital for each care type. (2) permits assignment only to hospitals that offer the service. (3) preserves existing capabilities. (4) limits the number of new designations. (5) enforces that at least fraction α of total demand is covered within the threshold T . This formulation balances efficiency (minimizing weighted distance) with equity (ensuring a minimum level of access for all).

4 Implementation

The model was implemented in Python using Gurobi 11.0 (professional license). Data processing used pandas and NumPy. For the base scenario we set $T = 60$ km, $\alpha = 0.6$, and $p^A = p^P = 10$. The model contains approximately 33,000 demand points and 634 hospitals, resulting in about 20 million binary assignment variables. Thanks to the powerful Gurobi solver and the model's structure, each instance solved in less than one second on a standard laptop.

We also conducted a sensitivity analysis by varying the coverage threshold $T \in \{30, 60, 90\}$ km and the number of new designations $p \in \{5, 10, 15, 20\}$. All model instances were solved to optimality.

5 Results

Table 1 presents the key performance metrics for all combinations of T and p . As expected, increasing p improves coverage and reduces travel distances. The coverage requirement of 60% ($\alpha = 0.6$) is met in all scenarios, but higher thresholds ($T = 30$) are more challenging and require more new centers to achieve comparable coverage.

Figure 1 illustrates the trade-off between the number of new centers and the average travel distance for both adult and pediatric care, aggregated across thresholds. The curves flatten after about 15 new centers, indicating diminishing returns.

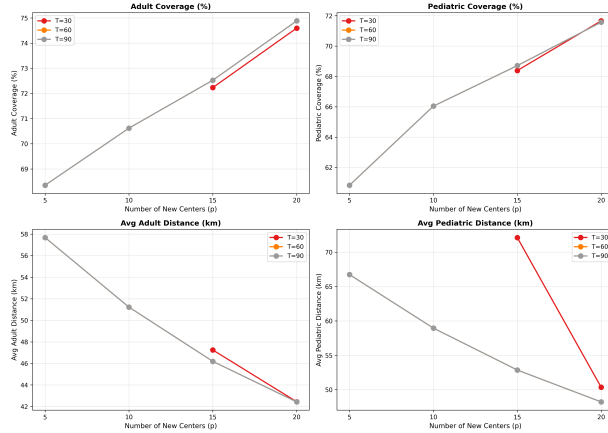


Figure 1: Average travel distance as a function of the number of new centers p . Adult and pediatric distances decrease rapidly initially and then level off.

The most striking improvement is in the maximum travel distance. With only five new centers, the maximum adult distance remains above 500 km, but with ten new centers it drops to around 430 km, and with fifteen it falls below 370 km. Pediatric maximum distance, initially very high for $T = 30$ (over 2300 km), is reduced to under 430 km with 20 new centers. This demonstrates the model’s ability to target underserved areas and substantially improve equity.

6 Sensitivity Analysis

The results in Table 1 reveal several important patterns:

- For the strictest threshold $T = 30$ km, even 15 new centers achieve only 72% adult coverage, and pediatric coverage lags behind. Increasing to 20 new centers boosts pediatric coverage by over three percentage points and cuts average pediatric distance by 30%.
- For $T = 60$ and $T = 90$, the coverage values are identical because the coverage constraint ($\alpha = 0.6$) is already satisfied; the model focuses on minimizing distance. Hence, the average and maximum distances are the same for both thresholds when p is fixed.
- The largest gains occur between $p = 5$ and $p = 10$. After $p = 15$, further improvements are modest. This suggests that a prudent policy would be to start with 10–15 new centers, achieving most of the potential benefit.

7 Conclusion

Our optimization model provides a rigorous, data-driven method for expanding burn care capacity in the United States. The results demonstrate that targeting a small number of hospitals for upgrading can lead to substantial improvements in access and equity. The sensitivity analysis quantifies the trade-offs between the number of new centers, coverage, and travel distances, enabling decision makers to choose an expansion plan that aligns with budget and policy goals. Future work could extend the analysis to include travel time via road networks, incorporate burn severity stratification, and evaluate the cost-effectiveness of different expansion strategies.

References

- [1] Louick EA, Phelan HA, Phillips BD, et al. (2024). Development of the national

Table 1: Sensitivity analysis: coverage and travel distances for different thresholds T (km) and numbers of new centers p .

T	p	Adult Cov (%)	Ped Cov (%)	Avg Adult (km)	Avg Ped (km)	Max Adult (km)	Max Ped (km)
30	15	72.2	68.4	47.2	72.1	369	2332
30	20	74.6	71.7	42.4	50.4	369	428
60	5	68.4	60.8	57.7	66.7	505	505
60	10	70.6	66.0	51.2	59.0	428	428
60	15	72.5	68.7	46.2	52.9	369	379
60	20	74.9	71.6	42.4	48.2	369	379
90	5	68.4	60.8	57.7	66.7	505	505
90	10	70.6	66.0	51.2	59.0	428	428
90	15	72.5	68.7	46.2	52.9	369	379
90	20	74.9	71.6	42.4	48.2	369	379

injury resource database (NIRD). *Burns*, 50(2):315–320.

- [2] Ivanko A, Garbuzov AE, Schoen JE, et al. (2024). The Burden of Burns: An Analysis of Public Health Measures. *Journal of Burn Care & Research*, forthcoming.
- [3] Carmichael H, Wiktor AJ, McIntyre RC, et al. (2019). Regional disparities in access to verified burn center care in the United States. *J Trauma Acute Care Surg*, 87(1):111–116.